

Extended State Observer Based Model Prediction Control for PMSM Field Weakening Control

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Abstract—This paper presents an extended state observer (ESO) based model predictive control (MPC) strategy for regulating the speed of permanent magnet synchronous motor (PMSM) during field weakening control, aiming to optimize the field weakening control performance in the presence of external disturbances and model uncertainty. An extended state observer is utilized to estimate disturbances caused by external disturbances and model parameter uncertainties, as well as nonlinear terms in the equations that are not easily handled. Through the incorporation of disturbance estimation and the anticipation of future disturbances within the prediction horizon, a predictive control law is derived by solving an optimization problem with constraints. Simulation results indicate that the proposed field weakening controller achieves better disturbance rejection, and solve the current and voltage constraints well.

Keywords—permanent magnet synchronous motor, field weakening, model predictive control, extended state observer

I. INTRODUCTION

Permanent magnet synchronous motor (PMSM) has gained extensive utilization across diverse domains like industrial servo drives, electric vehicles, and wind turbines, owing to its attributes of high power density, efficiency, and uncomplicated structure [1]. A key attribute of contemporary PMSMs is their capability to function within the field weakening region, effectively improving the torque characteristics of the motor. This extension proves pivotal in generating elevated torque at speeds surpassing the rated speed [2]. Viewed from a user standpoint, this capability translates into heightened productivity, rendering field weakening control an increasingly pivotal focus of research. Differing from slow speed control methods, the challenge of field weakening control involves the consideration of current and voltage constraints. Consequently, numerous methods have been proposed to solve this issue. Many classical approaches to PMSM field weakening control are based on cascaded field-oriented vector control (FOC), such as the formula-based approach [3], look-up table method [4], and negative current compensation method [5].

In recent years, in pursuit of enhanced performance, model predictive control (MPC) has been integrated to field weakening control. As advanced control technology, its primary concept is to utilize the dynamic model to forecast the future states and calculate the most optimal control action for the next moment by optimizing the target cost function [6][7]. MPC method is applied well to field weakening control owing

to the following reason that it is crucial to handle current and voltage constraints, and MPC can solve them effectively. Although numerous papers employ MPC for developing field weakening controllers, there exists many differences in their control objectives and structures. Single-loop MPC designs for field weakening control are adopted by [8] and [9], allowing for further improvement in the dynamic performance of the system, due to the absence of cascade bandwidth limitation [10]. Cascaded control structure is still utilized by [11] and [12], with the former employing MPC in current control and the latter applying MPC in speed control to optimize electromagnetic torque. The control system structured in cascade configuration has the capability to mitigate motor speed ripple and bolster system robustness, meanwhile achieve more refined control, so that each loop can focus on specific control task [13][14].

In field weakening control, in addition to the model's nonlinearity, there are nonlinearities present within current and voltage constraints, complicating the design of the controller, which is an obstacle in employing MPC within PMSM [15][16]. Given the computational power of prevailing industry-standard hardware, opting for linear MPC emerges as the most viable choice for realizing feasible implementations of control systems with rapid dynamics [17]. Treating all nonlinear terms as measured disturbances and using polygon constraints to approximate nonlinear constraints is done in [8]. However, the behavior of treating nonlinear terms as measured disturbances can limit the achievable control performance. Second order Runge-Kutta linearization method is employed in [9] to implement a single-loop MPC, and this paper utilizes tangent constraints to approximate nonlinear constraints, improving the control performance and enable the system to obtain a larger current and voltage output range than method in [8]. The above-mentioned control methods focus solely on applying MPC to field weakening control, but not account for disturbances from external disturbances and model uncertainty.

In this paper, in order to reduce the influence of external disturbances and model uncertainties, this paper proposes an MPC speed controller scheme based on extended state observer (ESO). Given the introduction of additional nonlinear terms by the interior permanent magnet synchronous motor (IPMSM) model, for the purpose of simplifying the controller design, the proposed ultimately adopts a cascaded structure, utilizing MPC for speed controller design solely. Initially, the proposed designs an extended state observer to estimate the lumped disturbance

caused by external disturbances and parameter uncertainties, as well as nonlinear terms in the equations that are not easily handled. Subsequently, the MPC is formulated using the motor model, which is established by incorporating the disturbance estimate into the PMSM system's dynamic model. Moreover, constraint inequalities are designed based on the current limit circle and voltage limit circle. Ultimately, optimal control inputs are derived by resolving Quadratic Programming problems while adhering to these constraints.

II. MODEL OF PMSM SERVO SYSTEM

In the d-q rotating reference frame, the mathematical model of the PMSM can be represented as following form (1):

$$\begin{cases} \frac{di_d}{dt} = -\frac{R_s}{L_d}i_d + n_p\omega\frac{L_q}{L_d}i_q + \frac{u_d}{L_d} \\ \frac{di_q}{dt} = -\frac{R_s}{L_q}i_q - \frac{L_d}{L_q}n_p\omega i_d - \frac{\psi_f}{L_q}n_p\omega + \frac{u_q}{L_q} \\ \frac{d\omega}{dt} = \frac{3n_p}{2J}[\psi_f i_q + (L_d - L_q)i_d i_q] - \frac{B_f\omega}{J} - \frac{1}{J}T_L \end{cases} \quad (1)$$

where R_s the stator resistance, L_d the direct-axis inductance, L_q the quadrature-axis inductance, ω the mechanical speed, n_p the number of poles, ψ_f the permanent magnet flux linkage, J the rotor inertia, B_f the friction coefficient, i_d and i_q the d-axis and q-axis currents, u_d and u_q the d-axis and q-axis voltages and T_L the load torque.

Field weakening control involves the consideration of current and voltage constraints. The maximum phase voltage limit U_m of the inverter is determined by the DC voltage, whereas the current limit I_m is determined by the motor current limit, and the following must be satisfied:

$$\begin{cases} i_d^2 + i_q^2 \leq I_m^2 \\ u_d^2 + u_q^2 \leq U_m^2 \end{cases} \quad (2)$$

A field weakening speed controller has been designed in this paper. The core idea is to leverage the advantages of MPC controllers in handling multi-input systems and constraint problem resolution, computing the optimal values of i_d^* and i_q^* , and utilize ESO to estimate lumped disturbance. The designed controller is illustrated in Fig. 1.

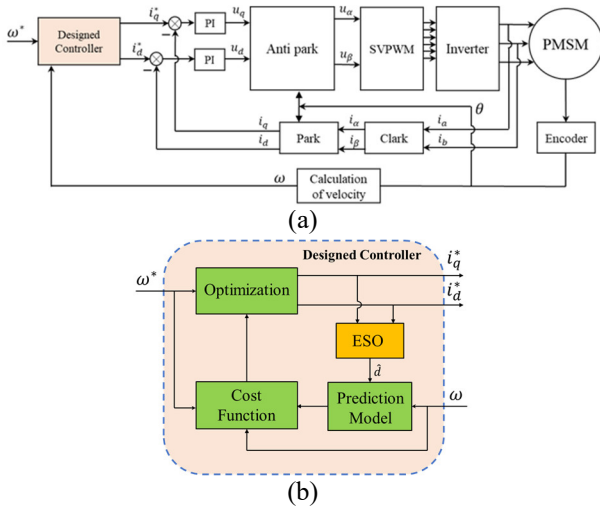


Fig. 1. Control block diagram and the designed controller. (a) Control block diagram. (b) Designed controller.

III. CONTROLLER DESIGN

A. Discrete Model of PMSM

The modified speed equation is shown as (3):

$$\frac{d\omega}{dt} = \frac{3n_p}{2J_m}[\psi_{fm}i_q^* + (L_{dm} - L_{qm})i_d^*i_q^*] - \frac{B_{fm}\omega}{J_m} + d_1 \quad (3)$$

where d_1 is defined as the lumped disturbance caused by external load variations, model uncertainty and current loop control errors. i_d^* and i_q^* are the outputs of speed controller. J_m , ψ_{fm} , B_{fm} , L_{dm} and L_{qm} are the nominal values of the mentioned motor parameters.

$$d_1 = -\frac{1}{J_m}\left[\frac{3n_p}{2}\psi_{fm}(i_q - i_q^*) - \frac{3n_p}{2}\Delta\psi_f i_q + \frac{3n_p}{2}(L_{dm} - L_{qm})(i_d i_q - i_d^* i_q^*) - \frac{3n_p}{2}(\Delta L_{dm} - \Delta L_{qm})i_d i_q + T_L + \Delta B_f \omega + \Delta J \frac{d\omega}{dt}\right] \quad (4)$$

where:

$$\begin{cases} \Delta J = J - J_m \\ \Delta\psi_f = \psi_f - \psi_{fm} \\ \Delta B_f = B_f - B_{fm} \\ \Delta L_d = L_d - L_{dm} \\ \Delta L_q = L_q - L_{qm} \end{cases} \quad (5)$$

Discretize (3) as follows:

$$\begin{aligned} \omega(k+1) &= \left(1 - \frac{B_{fm}T}{J_m}\right)\omega(k) + \frac{3n_p T}{2J_m}\psi_{fm}i_q^*(k) \\ &+ \frac{3n_p T}{2J_m}(L_{dm} - L_{qm})i_d^*(k)i_q^*(k) + Td_1(k) \end{aligned} \quad (6)$$

the incremental model is computed as:

$$\begin{aligned} \Delta\omega(k+1) &= \left(1 - \frac{B_{fm}T}{J_m}\right)\Delta\omega(k) \\ &+ \frac{3n_p T}{2J_m}[\psi_{fm} + (L_{dm} - L_{qm})i_d^*(k-1)]\Delta i_q^*(k) \\ &+ \frac{3n_p T}{2J_m}(L_{dm} - L_{qm})i_q^*(k-1)\Delta i_d^*(k) \\ &+ \frac{3n_p T}{2J_m}(L_{dm} - L_{qm})\Delta i_d^*(k)\Delta i_q^*(k) + T\Delta d_1(k) \end{aligned} \quad (7)$$

where T is the control period, and additionally:

$$\begin{cases} \Delta\omega(k) = \omega(k) - \omega(k-1) \\ \Delta i_q^*(k) = i_q^*(k) - i_q^*(k-1) \\ \Delta i_d^*(k) = i_d^*(k) - i_d^*(k-1) \\ \Delta d_1(k) = d_1(k) - d_1(k-1) \end{cases} \quad (8)$$

(9) is obtained by combining (6) and (7), $d(k)$ is the disturbance caused by the model uncertainties, external load, and model's nonlinearity:

$$\begin{cases} x(k+1) = A_d x(k) + B_{du} u(k) + B_{dd} d(k) \\ y(k+1) = C_d x(k+1) \end{cases} \quad (9)$$

where: $x(k) = [\Delta\omega(k) \ \omega(k)]^T$, $u(k) = [\Delta i_q^*(k) \ \Delta i_d^*(k)]^T$,

$$d(k) = Td_1(k) + \frac{3n_p}{2J_m}(L_{dm} - L_{qm})\Delta i_d^*(k)\Delta i_q^*(k),$$

$$A_d = \begin{bmatrix} 1 - \frac{B_{fm}T}{J_m} & 0 \\ 1 - \frac{B_{fm}T}{J_m} & 1 \end{bmatrix}, B_{du} = \frac{3n_p T}{2J_m} \begin{bmatrix} \psi_{fm} + (L_{dm} - L_{qm})i_d^*(k-1) & (L_{dm} - L_{qm})i_q^*(k-1) \\ \psi_{fm} + (L_{dm} - L_{qm})i_d^*(k-1) & (L_{dm} - L_{qm})i_q^*(k-1) \end{bmatrix},$$

$$B_{dd} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, C_d = [0 \ 1].$$

B. ESO Design

In order to enhance the disturbance rejection performance of the PMSM servo system, the proposed introduce ESO to estimate the lumped disturbance $d(k)$.

Firstly, modify (7) to (10):

$$\Delta\omega(k+1) = (1 - \frac{B_{fm}T}{J_m})\Delta\omega(k) + T_e^*(k) + d(k) \quad (10)$$

where $T_e^*(k) = \frac{3n_p T}{2J_m} \{ \psi_{fm} \Delta i_q^*(k) + (L_{dm} - L_{qm}) [\dot{i}_d^*(k-1) \Delta i_q^*(k) + \dot{i}_q^*(k-1) \Delta i_d^*(k)] \}$.

Due to the brief sampling period within the motor control system, the gradual changes in load torque and motor parameters in practical conditions result in the consideration that disturbances change slowly. According to the method proposed in [18] and [19], the discrete-time ESO is designed in the following form:

$$\begin{cases} \hat{z}(k) = a\hat{z}(k-1) + \hat{d}(k-1) + u(k-1) - \beta_1(\hat{z}(k) - z(k)) \\ \hat{d}(k) = \hat{d}(k-1) - \beta_2(\hat{z}(k) - z(k)) \end{cases} \quad (11)$$

where $a = 1 - \frac{B_{fm}T}{J_m}$, $z = \Delta\omega$, $u = T_e^*$, $\hat{d}$ is the estimated value of disturbance d , β_1 and β_2 are the observer gains. Modifying gains β_1 and β_2 enables the observer to converge.

C. Speed-Loop MPC Design

1) Prediction Model

The prediction model can be described as (12) by substituting the disturbance estimation $\hat{d}(k)$ to (9):

$$\begin{cases} x(k+1) = A_d x(k) + B_{du} u(k) + B_{dd} \hat{d}(k) \\ y(k+1) = C_d x(k+1) \end{cases} \quad (12)$$

For a clearer representation of the prediction process, we can define the prediction output Y and control input U in the form as shown in (13):

$$\begin{aligned} U &= [\Delta i_q^*(k) \quad \Delta i_d^*(k) \quad \Delta i_q^*(k+1) \quad \Delta i_d^*(k+1) \\ &\quad \dots \quad \Delta i_q^*(k+N_c-1) \quad \Delta i_d^*(k+N_c-1)]^T \\ Y &= [\omega(k+1|k) \quad \omega(k+2|k) \quad \dots \quad \omega(k+N_p|k)]^T \end{aligned} \quad (13)$$

where $U \in \mathbb{R}^{2N_c}$, $Y \in \mathbb{R}^{N_p}$, N_c is the control horizon, N_p is the prediction horizon, and $N_c \leq N_p$ [20]. Secondly, assume that the control variables remain constant after N_c steps of prediction, with disturbance estimate values replacing actual disturbances, and each step's disturbance is considered constant within the future prediction range. These assumptions are formulated as follows:

$$\begin{aligned} \dot{i}_q^*(k+i) &= \dot{i}_q^*(k+N_c-1), \quad i = N_c, \dots, N_p-1 \\ \dot{i}_d^*(k+j) &= \dot{i}_d^*(k+N_c-1), \quad j = N_c, \dots, N_p-1. \\ \hat{d}(k+l) &= \hat{d}(k), \quad l = 1, \dots, N_p-1 \end{aligned} \quad (14)$$

Therefore, based on the above definitions and assumptions, the predicted outputs at every sampling instance can be derived from (12).

First step:

$$\begin{cases} x(k+1|k) = A_d x(k) + B_{du} u(k) + B_{dd} \hat{d}(k) \\ y(k+1|k) = C_d x(k+1|k) \end{cases} \quad (15)$$

second step:

$$\begin{cases} x(k+2|k) = A_d^2 x(k) + A_d B_{du} u(k) + B_{du} u(k+1) \\ \quad + (A_d B_{dd} + B_{dd}) \hat{d}(k) \\ y(k+2|k) = C_d x(k+2|k) \end{cases} \quad (16)$$

N_p step:

$$\begin{cases} x(k+N_p|k) = A_d^{N_p} x(k) + A_d^{N_p-1} B_{du} u(k) + \dots \\ \quad + \sum_{l=0}^{N_p-N_c} A_d^l B_{du} u(k+N_c-1) + \sum_{l=0}^{N_p-1} A_d^l B_{dd} \hat{d}(k) \\ y(k+N_p|k) = C_d x(k+N_p|k) \end{cases} \quad (17)$$

based on the definition in (13), the predicted output sequence can be expressed in matrix format as:

$$Y = Fx(k) + \Phi U + M\hat{d}(k) \quad (18)$$

where:

$$\begin{aligned} F &= \begin{bmatrix} C_d A_d \\ C_d A_d^2 \\ \dots \\ C_d A_d^{N_p} \end{bmatrix}, M = \begin{bmatrix} C_d B_{dd} \\ C_d (I + A_d) B_{dd} \\ \dots \\ C_d \sum_{l=0}^{N_p-1} A_d^l B_{dd} \end{bmatrix}, \\ \Phi &= \begin{bmatrix} C_d B_{du} & 0 & \dots & 0 \\ C_d A_d B_{du} & C_d B_{du} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ C_d A_d^{N_p-1} B_{du} & C_d A_d^{N_p-2} B_{du} & \dots & C_d \sum_{i=0}^{N_p-N_c} A_d^i B_{du} \end{bmatrix}, \\ F &\in \mathbb{R}^{N_p \times 2}, M \in \mathbb{R}^{N_p}, \Phi \in \mathbb{R}^{N_p \times 2N_c}. \end{aligned} \quad (19)$$

2) Cost Function

The objective of MPC is to devise an optimal control sequence aimed at minimizing the error between the given reference sequence Y_r and the predicted output sequence Y , where:

$$Y_r = [1 \quad \dots \quad 1]^T \omega_r \quad (20)$$

thus, the cost function J is constructed as in (21):

$$J = (Y_r - Y)^T Q (Y_r - Y) + U^T R U \quad (21)$$

where Q and R are coefficient weighting matrices for the output error and control inputs, respectively. Generally, they can take the form of diagonal matrices as follows:

$$\begin{aligned} Q &= \text{diag}(q_1 \quad \dots \quad q_{N_p}) \\ R &= \text{diag}(r_1 \quad \dots \quad r_{N_c}) \end{aligned} \quad (22)$$

3) Constraint Solving

After defining the cost function J , the next step involves calculating the optimal control input sequence by minimizing this cost function. During this process, it is essential to consider the current and voltage constraints involved in field weakening control. Therefore, this task is transformed into a Quadratic Programming problem with constraints, which can be effectively addressed by applying the Lagrange dual principle [21].

First, J can be updated as (23) by combining (18) and (21):

$$\begin{aligned} J &= \Theta + U^T (\Phi^T Q \Phi + R) U \\ &\quad - 2U^T \Phi^T Q [Y_r - Fx(k) - M\hat{d}(k)] \end{aligned} \quad (23)$$

where Θ is independent of the input U .

The next step is to design the constraint conditions, which is a critical component of field weakening control. The constraint conditions can be designed based on the current and voltage limit circle involved in field weakening control [22],

which includes two nonlinear inequality constraints as shown in (2). According to the article [22], the constraint (2) can be rewritten as (24):

$$\begin{cases} i_d^2 + i_q^2 \leq I_m^2 \\ (L_d i_d + \psi_f)^2 + (L_q i_q)^2 \leq U_{om}^2 \end{cases} \quad (24)$$

where $U_{om} = (U_m - RI_m)/|n_p \omega|$, which decreases as the speed increases.

First, provide the four sets of constraints in (25), treating the two circles as two externally tangent rectangles:

$$\begin{cases} -I_m \leq i_d^*(k) \leq I_m \\ -I_m \leq i_q^*(k) \leq I_m \\ \frac{-U_{om} - \psi_f}{L_d} \leq i_d^*(k) \leq \frac{U_{om} - \psi_f}{L_d} \\ \frac{-U_{om}}{L_q} \leq i_q^*(k) \leq \frac{U_{om}}{L_q} \end{cases} \quad (25)$$

Subsequently, utilizing additional boundary information from the limit circles, the nonlinear constraints are simplified into linear constraints as shown in the graph:

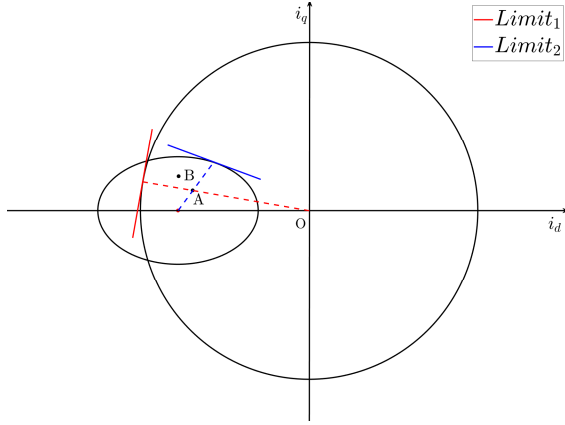


Fig. 2. Circle tangential linearization [12].

In Fig. 2, point A represents the current $(i_d(k), i_q(k))$ at time step k , and point B represents the control input $(i_d^*(k), i_q^*(k))$ at time step k . Drawing a line through points O and A that intersects the current limit circle and then finding the tangent line through the intersection point provides constraint $Limit_1$, as similarly derived from the voltage limit circle to obtain constraint $Limit_2$. By adding these two constraints, it can be ensured that the control input B at time step k falls within the two limit circles. Even if at some point it falls outside the limit circle, due to the faster control frequency, it can be quickly corrected to fall within the constraint range. Moreover, as the controlled object is a motor, it is typically designed to withstand short-term overcurrent, and short-term overcurrent does not typically pose a serious safety issue [12]. The resulting constraint formulation is simple and can be expressed as in (26):

$$\begin{aligned} Limit_1 : & i_d(k) i_d^*(k) + i_q(k) i_q^*(k) \leq I_m \sqrt{i_d^2(k) + i_q^2(k)} \\ Limit_2 : & a L_d i_d^*(k) + b L_q i_q^*(k) \leq U_{om} \sqrt{a^2 + b^2} - a \psi_f \end{aligned} \quad (26)$$

where $a = L_d i_d(k) + \psi_f$, $b = L_q i_q(k)$.

Based on definitions (12) and (13), rewrite constraints (25) and (26) in matrix form:

$$WU \leq \gamma \quad (27)$$

where:

$$W = \begin{bmatrix} 0 & 1 & \dots & 0 \\ 0 & -1 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ -1 & 0 & \dots & 0 \\ i_q(k) & i_d(k) & \dots & 0 \\ a L_d & b L_q & \dots & 0 \end{bmatrix}, \gamma = \begin{bmatrix} \min\left(I_m - i_d^*(k-1), \frac{U_{om} - \psi_f}{L_d} - i_d^*(k-1)\right) \\ \min\left(I_m + i_d^*(k-1), \frac{U_{om} + \psi_f}{L_d} + i_d^*(k-1)\right) \\ \min\left(I_m - i_q^*(k-1), \frac{U_{om}}{L_q} - i_q^*(k-1)\right) \\ \min\left(I_m + i_q^*(k-1), \frac{U_{om}}{L_q} + i_q^*(k-1)\right) \\ I_m \sqrt{i_d^2(k) + i_q^2(k)} - i_d^*(k-1) i_d(k) - i_q^*(k-1) i_q(k) \\ U_{om} \sqrt{a^2 + b^2} - a \psi_f - a L_d i_d^*(k-1) - b L_q i_q^*(k-1) \end{bmatrix} \quad (28)$$

$W \in \mathbb{R}^{6 \times 2N_c}, \gamma \in \mathbb{R}^6$.

Combining (23) and (28), the resolution for the optimal control sequence can be reformulated as the solution to the subsequent constrained Quadratic Programming problem:

$$\begin{aligned} \min_U J &= \frac{1}{2} U^T E U + U^T G \\ s.t. & WU \leq \gamma \end{aligned} \quad (29)$$

where $E = 2(\Phi^T Q \Phi + R)$, $G = -2\Phi^T Q[Y_r - Fx(k) - M\dot{a}(k)]$.

IV. SIMULATION RESULTS

In order to substantiate the efficacy of the method delineated in this paper, a simulation of a PMSM servo system was conducted. To verify the proposed strategy subject to model uncertainties, the parameter modeling errors were set to approximately 10% of the system's nominal values. The simulation compared two different field weakening control methods: an MPC field weakening speed controller and an MPC+ESO field weakening speed controller. The parameters of the PMSM are as shown in Table I.

TABLE I. PARAMETERS OF THE PMSM

Parameter	Nominal Value	Actual Value
Sator resistance, R_s	1.6Ω	1.735Ω
d - axis inductance, L_d	4.5mH	5mH
q - axis inductance, L_q	5.5mH	6mH
Rated flux linkage, ψ_f	0.05Wb	0.056Wb
Pole pairs, n_p	5	5
Maximum voltage, U_m	200V	200V
Maximum current, I_m	12A	12A

For the sake of a fair comparison, the current loop PI parameters for all two control methods were set to the same values. The prediction horizon $N_p = 4$, the control horizon $N_c = 3$, the pole of ESO -500, the speed loop control period T 2ms, and the current loop control period 0.2ms. The target speed reference value was set at 12000rpm. At time 0.2s, a load torque T_L of 0.5 N·m was introduced to evaluate system performance under external disturbances.

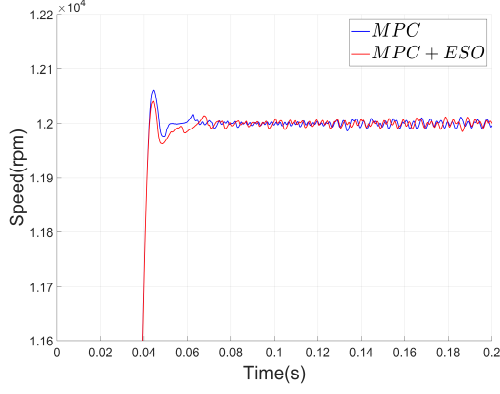


Fig. 3. Speed response at 12000rpm speed reference.

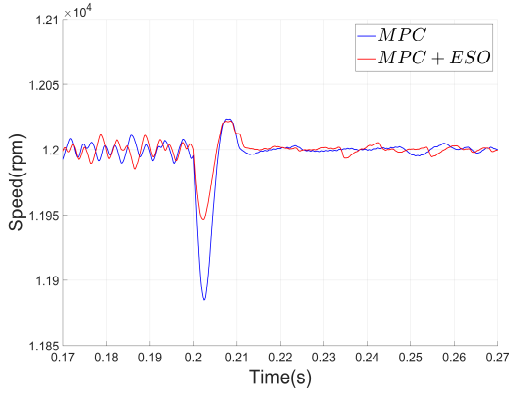


Fig. 4. Speed response with load torque.

Fig. 3 and Fig. 4 depict the speed response at target speed of 12000rpm. Fig. 3 compares the dynamic performance of two speed controllers. Fig. 4 compares the disturbance rejection of two speed controllers. Table II gives the quantitative data comparing the performance of the two cases. It can be observed from Table II that MPC+ESO has a smaller overshoot during startup and reflects better disturbance rejection during sudden load.

TABLE II. CONTROL PERFORMANCE

Controller	Rise time (0.5%)	Overshoot	Sudden load Decrease	Sudden load recovery time
MPC	42ms	0.6%	116rpm	18ms
MPC+ESO	42ms	0.4%	54rpm	14ms

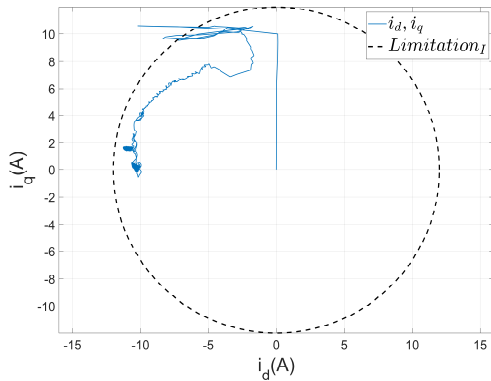


Fig. 5. Keeping the current constrains.

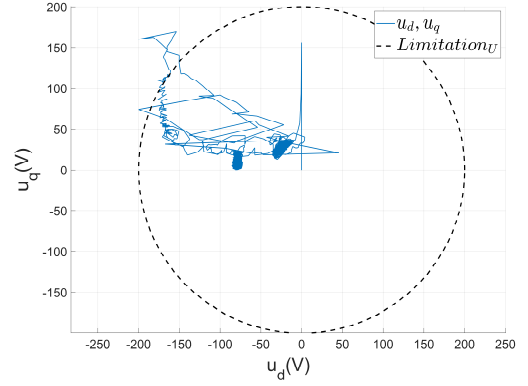


Fig. 6. Keeping the voltage constrains.

Fig. 5 and 6 show the current and voltage trajectories. It can be observed that most of points are located within the limit circles, essentially meeting the requirements for field weakening control.

V. CONCLUSION

In this paper, a field weakening control method for PMSM servo systems has been proposed, which combines ESO with MPC. The primary objective of this field weakening controller is to mitigate the impact of parameter uncertainties and external disturbances.

Initially, ESO has been utilized to estimate lumped disturbances resulting from external perturbations and uncertainties in model parameters. Subsequently, an MPC controller has been developed which considers disturbances, using the dynamic model of the PMSM. Moreover, constraint inequalities are designed based on the current limit circle and voltage limit circle. Finally, the optimal control inputs are computed by solving a Quadratic Programming problem with constraints.

While addressing current and voltage constraints, this controller has demonstrated better disturbance rejection, which has been verified by simulation.

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